

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

1. **LO 4.4.A** — Express the motion of a point around a circle parametrically by applying transformations to $(x(t), y(t)) = (\cos t, \sin t)$, and parametrize the segment between two given points using an initial position and constant rates of change. **see notes boxes below**

Vocabulary & Key Concepts

unit circle parametrization: The equations $x(t) = \cos t$, $y(t) = \sin t$, $0 \leq t \leq 2\pi$ model one full **counter-clockwise** revolution starting at $(1, 0)$ and centered at the **origin**.

radius / center: In $(h + r \cos t, k + r \sin t)$: r is the **radius** and (h, k) is the **center** of the circular path.

orientation: Using $(\cos t, \sin t)$ gives **counterclockwise** motion. Replacing t with $-t$ gives **clockwise** motion.

linear parametrization: The parametrization $x(t) = x_1 + (x_2 - x_1)t$, $y(t) = y_1 + (y_2 - y_1)t$, $0 \leq t \leq 1$, traces the segment from (x_1, y_1) to (x_2, y_2) .

Hook

Entry Question: A Ferris wheel has radius 30 ft with its center 35 ft above the ground. A rider boards at the rightmost seat.

Q1. What are the rider's starting coordinates (x, y) relative to a center at the origin?

$(30, 0)$ relative to the center; in ground coordinates, $(30, 35)$.

Q2. As the wheel turns counterclockwise, what do you expect to happen to the rider's height y ?

The height increases to a maximum of $35 + 30 = 65$ ft (at the top) then decreases back to $35 - 30 = 5$ ft (at the bottom), following a sine curve.

LO 4.4.A — Part 1: Unit Circle Parametrization (EK 4.4.A.1)

Definition: The parametric function $(x(t), y(t)) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$, traces the **unit** circle exactly **once**, moving **counterclockwise**.

Complete the table:

t	$x(t) = \cos t$	$y(t) = \sin t$	(x, y)
0	1	0	$(1, 0)$
$\pi/2$	0	1	$(0, 1)$
π	-1	0	$(-1, 0)$
$3\pi/2$	0	-1	$(0, -1)$
2π	1	0	$(1, 0)$

Teacher Note

Emphasize that the table values come directly from the unit-circle cosine/sine values from the warm-up. Students can verify geometrically by looking at the standard unit circle diagram.

LO 4.4.A — Part 2: Transformations of the Circle (EK 4.4.A.2)

General form:

$$x(t) = \mathbf{h} + \mathbf{r} \cos t, \quad y(t) = \mathbf{k} + \mathbf{r} \sin t, \quad 0 \leq t \leq 2\pi$$

- Center: $(\mathbf{h}, \mathbf{k})$ Radius: $\mathbf{r}$
- Counterclockwise when t increases / Clockwise: replace t with $-t$

Ferris Wheel Example. Radius 30 ft, center at $(0, 35)$, counterclockwise:

$$x(t) = 30 \cos t \quad y(t) = 35 + 30 \sin t$$

Evaluate at $t = \pi/2$: $x(\pi/2) = 0$, $y(\pi/2) = 65$

The rider is at position $(0, 65)$ after a quarter turn.

Clockwise version (replace t with $-t$):

$$x(t) = 30 \cos(-t) = 30 \cos t \quad y(t) = 35 + 30 \sin(-t) = 35 - 30 \sin t$$

Teacher Note

Common error: students negate *both* components for clockwise. Stress that $\cos(-t) = \cos t$ (cosine is even), so only sin changes sign.

LO 4.4.A — Part 3: Linear Parametrization (EK 4.4.A.3)

Segment from (x_1, y_1) to (x_2, y_2) :

$$x(t) = x_1 + (x_2 - x_1)t, \quad y(t) = y_1 + (y_2 - y_1)t, \quad 0 \leq t \leq 1$$

At $t = 0$: position is (x_1, y_1) . At $t = 1$: position is (x_2, y_2) . At $t = 0.5$: position is the **midpoint**.

Example: Segment from $(2, 5)$ to $(6, 1)$.

$$x(t) = 2 + (4)t = 2 + 4t \quad y(t) = 5 + (-4)t = 5 - 4t$$

Complete the table:

t	$x(t)$	$y(t)$	(x, y)
0	2	5	$(2, 5)$
0.5	4	3	$(4, 3)$
1	6	1	$(6, 1)$